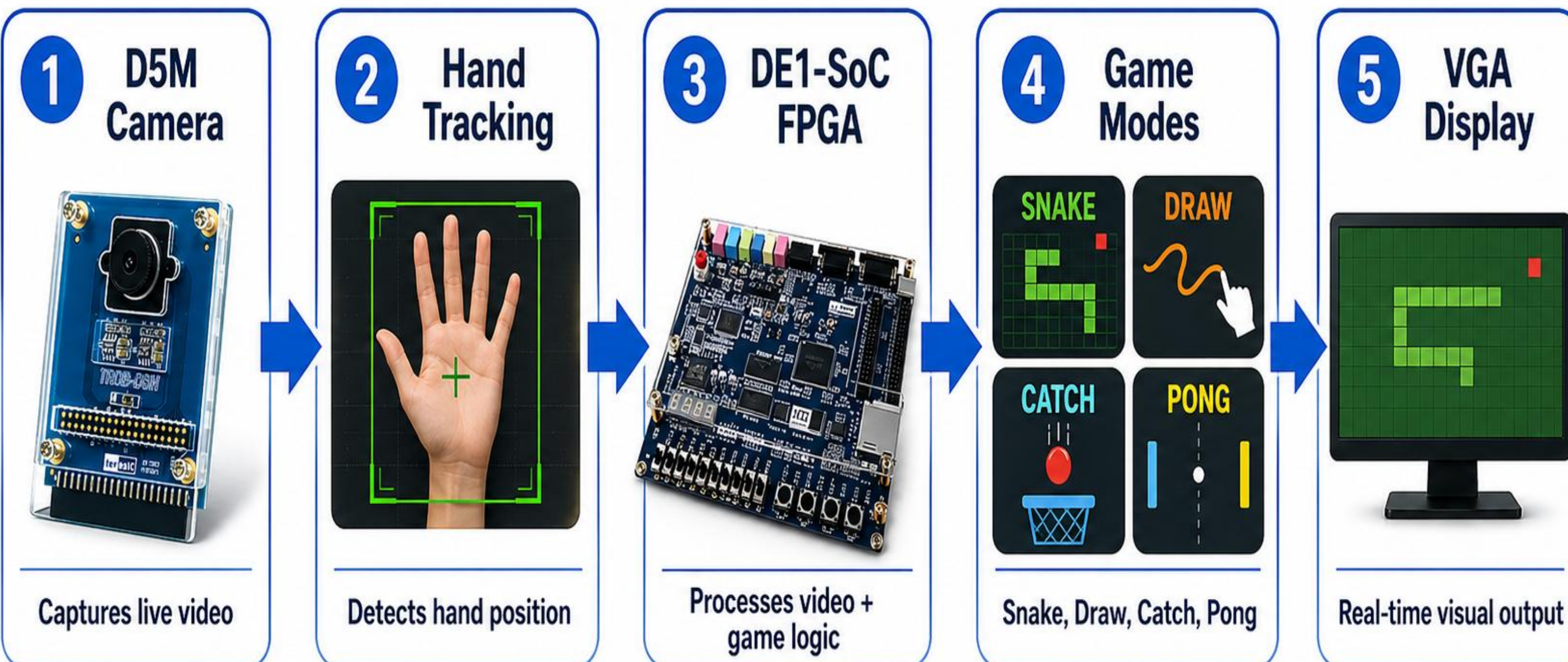
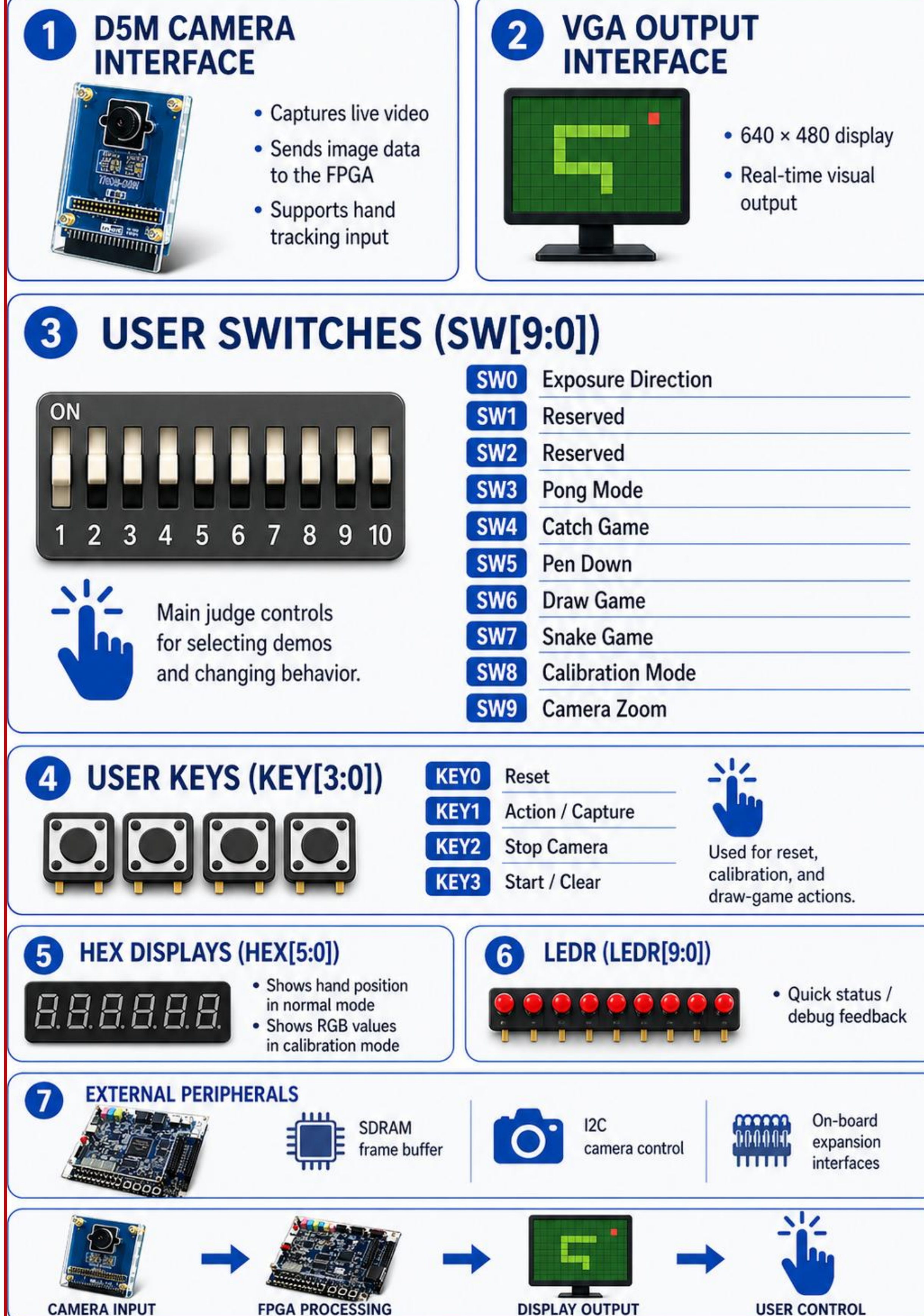


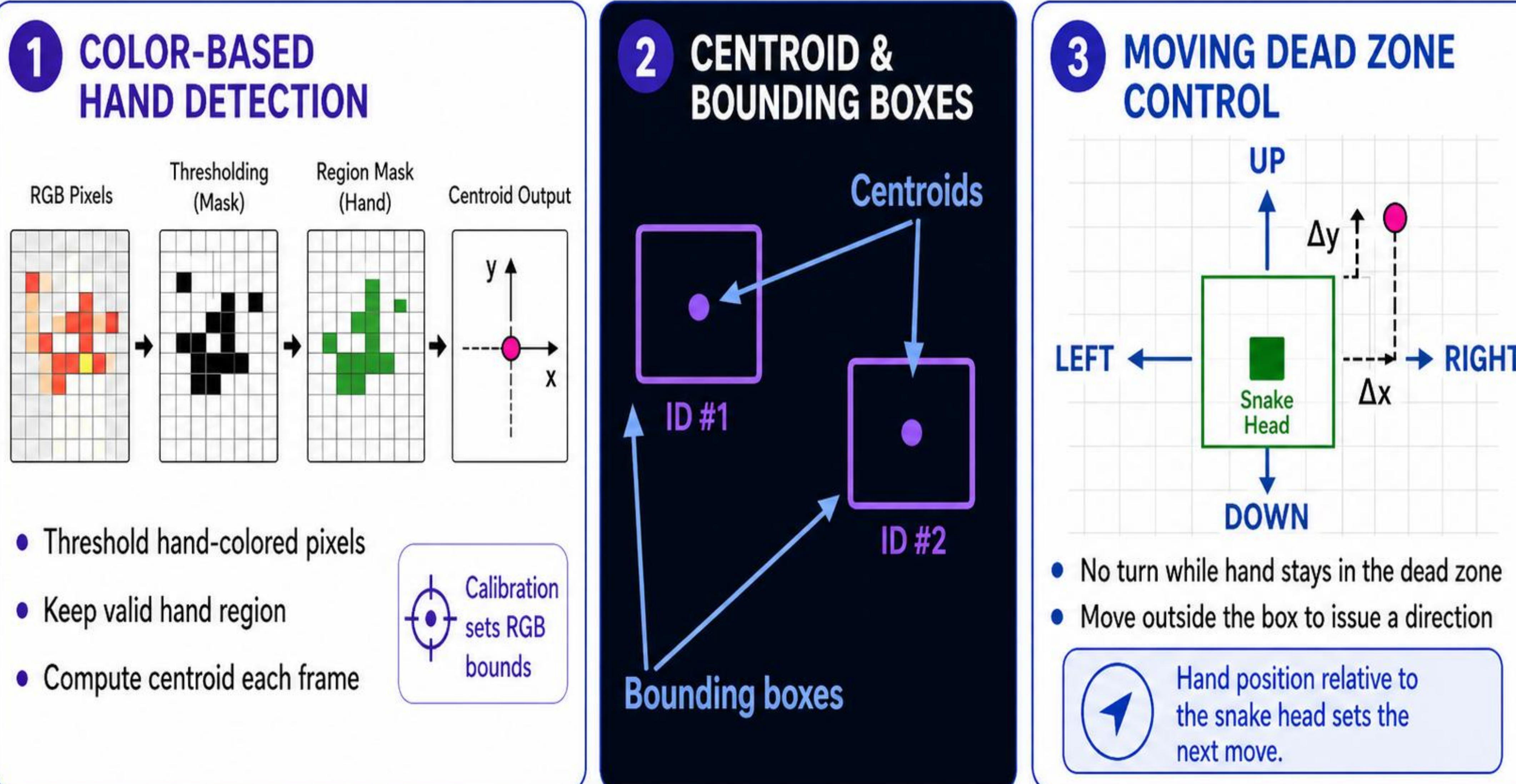
System Overview



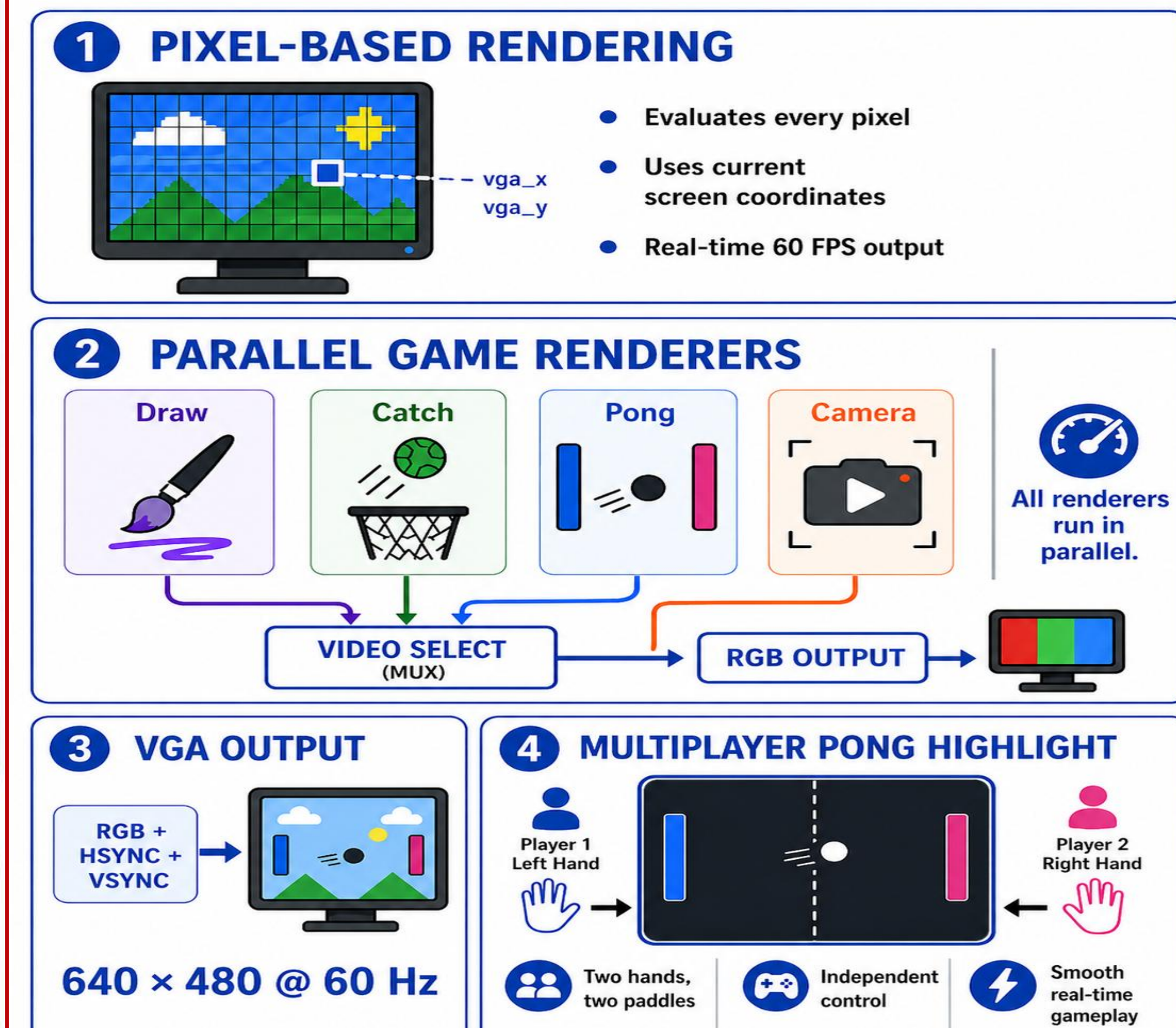
Hardware Interfaces & Peripherals



Core Algorithms



Rendering Pipeline & Game Output



Challenges

- Clock Domain Crossing (Camera → VGA):** Safe synchronization was required to transfer hand tracking data between different clock domains without instability.
- Latency vs. Responsiveness Tradeoff:** Frame-based centroid updates improved stability but introduced slight delay in control response.
- Designing an Intuitive Control Scheme:** A moving dead zone around the snake head was needed to reduce jitter and enable natural gameplay.

Future Considerations

- AI/ML-Based Hand Tracking:** Replacing RGB thresholding with ML techniques could improve robustness to lighting changes and enable gesture recognition.
- Enhanced Hardware Platform:** Upgrading to a more powerful FPGA or camera would allow for better performance and graphics.
- Expanded Gameplay:** Adding more advanced game mechanics, scoring systems, AI opponents, and modes would enhance overall user experience.